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Reg No.: _____

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S4 (R,S) / S4 (WP) (R) / S2 (PT) (S, FE) Examination May 2024 (2019 Scheme)

Course Code: EET 206

Course Name: DIGITAL ELECTRONICS

Max. Marks: 100

Duration: 3 Hours

PART A

(Answer all questions; each question carries 3 marks)

Marks

- | | | |
|----|---|-----|
| 1 | Perform the following number conversions. | (3) |
| | a) $(875.84)_{10}$ to hexadecimal b) Gray code 110010 to binary. | |
| 2 | What is meant by a universal gate? List two universal gates and write its truth table. | (3) |
| 3 | Prove that $AB + \bar{A}C = (A + C)(\bar{A} + B)$, where A, B , and C are logic variables. | (3) |
| 4 | State De Morgan's theorem. Simplify the Boolean expression $f(P, Q) = \overline{\overline{PQ} + \bar{P} + PQ}$ into minimum number of literals by applying De Morgan's theorem. | (3) |
| 5 | What is the advantage of a priority encoder over an encoder? | (3) |
| 6 | Draw the block diagram of a 4:1 multiplexer. Write down its function table and develop a Boolean expression for its output. | (3) |
| 7 | What are the asynchronous inputs to a flip-flop? Explain its function. | (3) |
| 8 | Compare the functioning of S-R flip-flop and J-K flip-flop with the help of truth tables. | (3) |
| 9 | Differentiate between Moore and Mealy state machines. | (3) |
| 10 | What is FPGA? What are the components of a typical FPGA logic block? | (3) |

PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

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|----|---|-----|
| 11 | a) Express $(-81.75)_{10}$ in binary using (i) 12-bit 2's complement form, and (ii) sign-magnitude form. | (6) |
| | b) Explain the working of TTL NAND gate with the help of an internal diagram. | (8) |
| 12 | a) Determine the range of 4 bit binary number which is expressed in (i) signed magnitude form (ii) 1's complement form (iii) 2's complement form. | (6) |
| | b) Explain the working of CMOS NOR gate with the help of an internal diagram. | (8) |

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Module -2

- 13 a) Expand the Boolean function, $f(A,B,C) = A\bar{B}C + \bar{C} + \bar{A}C$ into canonical SOP form. (4)
- b) Derive an expression for the sum and the carry of a full adder. Implement the derived sum and carry expressions using only NAND gates. (10)
- 14 a) Implement the Boolean function $f(X,Y,Z) = \sum m(1,2,3,4,5,7)$ using minimum number of NAND gates. (4)
- b) What is a carry look-ahead adder? Why is it preferred over parallel adders? Draw the logic circuit of a full adder which can generate look-ahead carry bit. (10)

Module -3

- 15 Explain the working of a comparator which can compare two 2-bit binary numbers, $A = A_2A_1$ and $B = B_2B_1$ (A_1 and B_1 are the LSB and A_2 and B_2 are MSB). Deduce expressions for the outputs when $A > B$, $A < B$, and $A = B$. Implement these expressions using logic gates. (14)
- 16 a) Implement the following sum of minterms expression using an 8:1 multiplexer. $f(A,B,C,D) = \sum m(0,2,5,7,12,13,14,15)$. (7)
- b) What is the significance of parity bit? Draw the logic circuit diagrams of an even parity bit generator and checker for a 4-bit data. (7)

Module -4

- 17 a) Draw the logic diagram of a 4-bit Johnson counter which employs D flip-flops. Write down its count sequence table and draw the timing diagram. (10)
- b) What is a shift register? List down any three applications of shift registers. (4)
- 18 Design and implement a mod-6 synchronous up-counter using J-K flip-flops. Draw the state diagram and relevant excitation table. Also, check for lock-out condition. (14)

Module -5

- 19 a) Write the Verilog code for AND gate, OR gate and half adder. (4)
- b) With the help of a neat circuit diagram, explain how a R-2R ladder type DAC converts a digital signal to analog signal. (10)
- 20 a) Differentiate between PAL and PLA. (4)
- b) Explain the working of a flash type analog to digital converter (ADC) with the help of a neat circuit diagram. (10)
